M.SC. PROJECT PRESENTATION · CUSAT

SAR-Guided Satellite Cloud Removal

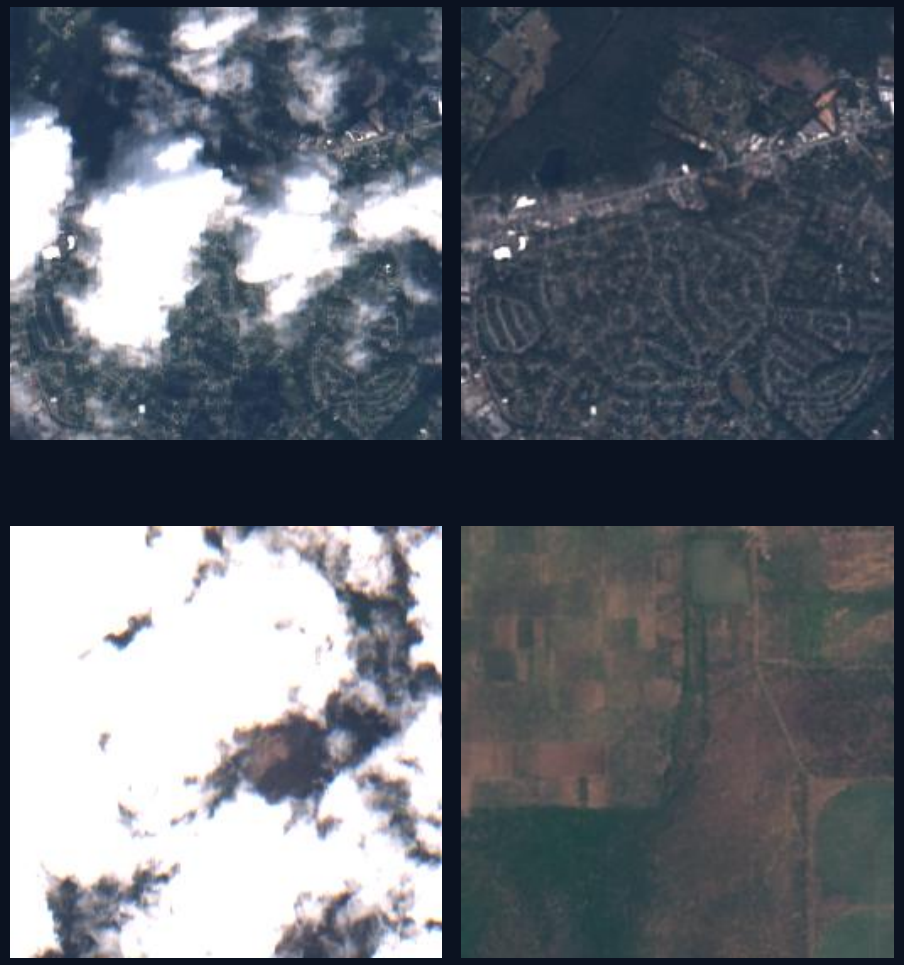
Seeing through clouds by combining radar with deep learning

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Cloudy input

Our output

OVERVIEW

What we will cover

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The problem

Why clouds are a big issue

02

The data

Satellites, dataset, preparation

03

The model

How ECRformer works

04

Sharper results

Fixing blur with smart losses

05

Results

What the model produces

06

Conclusion

Limits and takeaways

Clouds hide the ground

More than half of the Earth is covered by cloud at any moment.

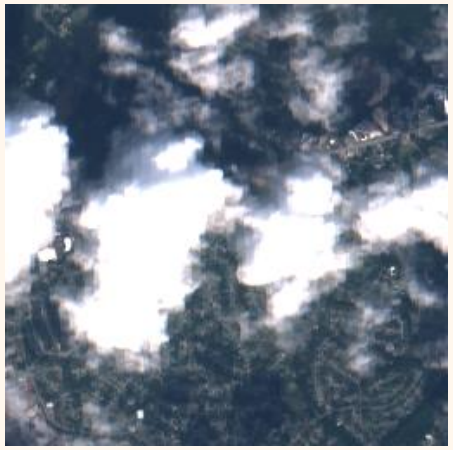
Optical satellites cannot see through them — the ground simply disappears.

Goal: rebuild the hidden ground so the image is usable again.



A real cloudy Sentinel-2 image

Two satellites, two strengths

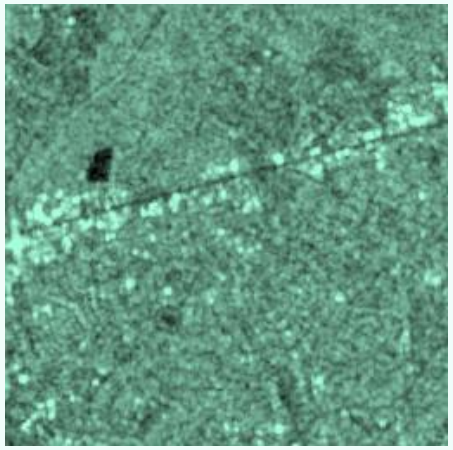


Sentinel-2

Optical camera

+ Rich natural colour
− Blocked by clouds

Sees sunlight reflected by the ground



Sentinel-1

SAR radar

+ Passes through clouds
− No natural colour

Sends its own microwaves and listens for echoes

Our idea → use radar structure to fill in what the clouds hide.

02 · THE DATA

SEN12MS-CR dataset

Paired radar, cloudy and cloud-free images of the same place

33,468
training patches

1,115
validation patches

256 × 256
pixels per patch · 10 m each

WHAT GOES INTO THE MODEL

4 optical bands + **2** radar bands + **1** cloud mask = **7** input channels

Preparing the data in 3 steps

1

Find the clouds

ESA's s2cloudless tool marks every cloudy pixel, giving a cloud mask.

→

2

Scale the radar

Radar values (in decibels) are rescaled to the range 0 – 1.

→

3

Stop shortcuts

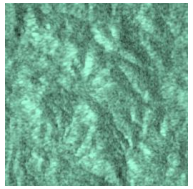
In 65% of training samples, cloudy areas are blurred so the model must rely on radar.



Cloudy



Cloud mask



Radar

How the system works

Inputs

Cloudy image
Radar · Cloud mask

→

ECRformer

Deep network that rebuilds the ground

→

Prediction

A full cloud-free guess of the scene

→

Blend

Keep real clear pixels, fill only the clouds

Training

Random 128 × 128 crops. A critic network (PatchGAN) judges if the result looks real.

Testing

Full 256 × 256 scenes. The critic is removed — only ECRformer runs.

Inside ECRformer

An hourglass-shaped network: shrink to understand, expand to rebuild

Two-door input

Radar and optical enter through separate layers first — they behave very differently.

Encoder

Shrinks the image step by step while learning richer features.

Attention core

Looks at the whole scene at once to connect distant clues.

Decoder

Expands back to full size. Skip links carry fine edges across.

Model size: about 8.3 million parameters — light enough for a single desktop GPU.

Train small, test big

Normal attention

Compares every pixel with every pixel.
Its size grows with the image — change the image size and it breaks.

ECRformer attention (MDTA)

Compares feature channels with each other.
Its size stays the same for any image size.

Result → trained on 128 × 128 crops, runs directly on 256 × 256 scenes.

The blur problem



Basic model (blurry)



Real clear image

Why does it blur?

1 Unsure what lies under the cloud, the model plays safe and paints an average.

2 Radar and optical images are often shifted by 1–2 pixels, so sharp edges get punished.

Teaching the model to be sharp

Seven loss functions, grouped by what they check

Colour

Is the colour right?

Scale-decoupled L1

Structure

Are the edges right?

VGG perceptual

Multi-scale gradient

Edge correlation

Local contrast

Realism

Does it look real?

PatchGAN adversarial

Feature matching

The PatchGAN critic

A second network that acts like a **strict art examiner.**

- Looks at small regions and scores each one real or fake.
- Blurry patches get low scores, pushing ECRformer to draw sharp detail.
- Checks cloudy areas 4× more strictly than clear ones.

Used only during training — removed at test time.



Illustration: each score judges one region
(red = looks blurry / fake)

Building the final image

$$\text{Output} = (1 - M) \times \text{Cloudy} + M \times \text{Prediction}$$

Clear sky (M = 0)

Real satellite pixels are kept exactly as they are.

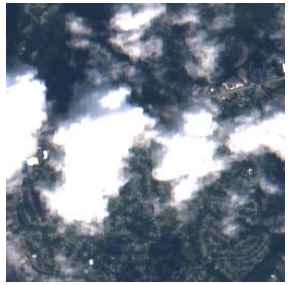
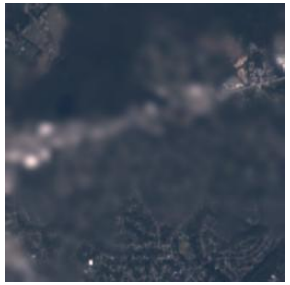
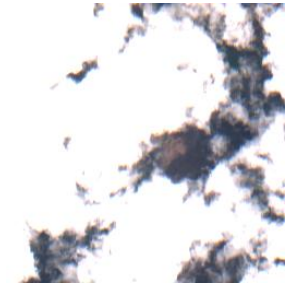
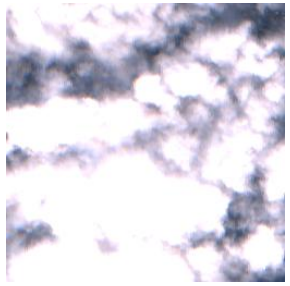
Thick cloud (M = 1)

The model's reconstruction fills the gap.

Cloud edges

Soft blending, so no visible seams.

Results on real scenes

	Cloudy input	Basic model	Our model	Real clear image
Suburban				
Farms				
Lake				

05 · RESULTS

What the numbers say

Compared with the basic model, across four cloud-cover groups

+47% to +133%

more fine detail recovered

≈ 0.9

detail ratio (1.0 = as sharp as reality)

30 – 32 dB

PSNR under the cloud

Basic model detail ratio: 0.39 – 0.63 → Ours: 0.90 – 0.93

Detail ratio = how much of the real image's fine texture is reproduced.

An honest look

The trade-off

- Sharper images score slightly lower on PSNR (0.3 – 1.2 dB).
- PSNR rewards safe, blurry averages — the known perception–distortion trade-off.
- We chose sharpness, which matters for real use.

Limitations

- Cloud shadows are not always caught by the cloud mask.
- The clear reference image is taken on a different day.
- Training too long adds artificial grain, so we keep the best checkpoint.

CONCLUSION

What we achieved

- Removed clouds by combining radar with optical images
- Turned blurry reconstructions into sharp, detailed ones
- Trained on small crops, works on full scenes



Before · After

Thank you

Questions are welcome

APPENDIX

Detailed pipeline (for reference)

